

Teacher notes

Enzyme Action: Preparing the pH Investigation

Staff planning and answers

Slide book directions and choices

Shared model · picture observation prompt

In your book: Glue in today's strip; date it. Answer the four clues.

Slide 3 · Starter · what can iodine tell us?

In your book: Choose an answer; give one reason using the evidence.

Slide 4 · I do · bind, break down, release

In your book: Watch the model. No writing yet.

Slide 5 · We do · label the enzyme model

In your book: Complete the shared task; record one decision and its reason.

Slide 6 · I do · connect pH to the measure

In your book: Follow the worked example; note the key reasoning step.

Slide 7 · We do · organise the investigation

In your book: Record one evidence claim and explain its limit.

Slide 8 · You do · prepare a valid comparison

In your book: Answer your chosen route using the supplied evidence.

Slide 9 · Review and improve

In your book: Improve one explanation; mark the change // and read it back.

Slide 10 · Exit · mechanism, measure, control

In your book: Answer both exit questions under your work.

P17 · SLT planning and evaluation

Lesson code: SCI-LAUNCH-A1-W08-L1. SoW reference: LAUNCH Science, Autumn 1 return block; L1.

Exact SoW outcome: I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

Source placement: enzyme activity enrichment consolidates Topic 1; the workbook has no separate Autumn 1 Week 8 row. This is the supplied return-period mapping, not a newly invented workbook objective.

This lesson: I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

Evidence (scheme of work): Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec). Evidence (this lesson): Use enzyme, substrate, active site and specificity accurately in an explanation.; Explain how an unsuitable pH can change an active site's shape and reduce enzyme activity.; Identify pH as the independent variable, time to starch disappearance as the dependent variable, and suitable control variables.

Evidence location: the pupil's explanation under the book strip, with route, actual support, review edit // and exit response. Keep raw observations distinct from illustrative data.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A1-W08-L1. Record the observed science and prompt level; clicks and access aids do not earn a marking code.

Stage	Minutes	Task	Staff move
Arrival task	4	Choose a route. Think first, then check the answers together.	At arrival, say the spoken line on the marking card once. Use the marking card's spoken arrival guidance once. Read the response with the help used in view. Make one same-lesson adjustment by about minute eight. IF NEW — 1 min read local facts; 2 min respond; 1 min sample and reveal. IF REVISION — 0.5 min orient; 2.5 min respond; 1 min sample and reveal.
Starter · what can iodine tell us?	4	Starch is detected in the sampled drop.	No reagents needed today. Read labels aloud: colour is not the only clue. Choose the arrival adjustment from the first-lesson-back card. IF NEW — 2 min show and name outcomes; 1 min compare; 1 min check endpoint. IF REVISION — 1 min inspect; 2 min justify outcomes; 1 min resolve the endpoint.
I do · bind, break down, release	5	The substrate has a complementary fit with the enzyme's active site.	Pause each step. This is a simplified shape model, not literal rigid molecules. Enzymes can flex; amylase is not a living organism. IF NEW — 3 min model all three moments; 1 min name terms; 1 min pupil retells. IF REVISION — 1 min orient; 3 min predict each step; 1 min repair an explanation.

Stage	Minutes	Task	Staff move
We do · label the enzyme model	5	Drag a card, or select it then choose its destination.	Named involvement: invite a pupil's label choice and engage with the explanation. Select-card then destination works with keyboard/touch; paper labels give the same science. IF NEW — 1 min read parts; 3 min match and explain; 1 min check. IF REVISION — 0.5 min orient; 3 min label and justify; 1.5 min compare.
I do · connect pH to the measure	4	pH can affect an enzyme's active site and activity.	Do not assume pH 7 is a universal optimum. Reduced activity alone does not prove permanent denaturation. A shorter endpoint time implies higher relative rate only for matched amounts/endpoints. IF NEW — 2 min connect pH and active site; 1 min link to time; 1 min check. IF REVISION — 1 min orient; 2 min explain causal chain; 1 min distinguish claim and evidence.
We do · organise the investigation	4	Use the arrows to organise the steps. Read the sequence aloud before checking.	F/B may swap if both precede mixing. This is planning, not permission to handle reagents. The later lesson needs the centre method and risk assessment. IF NEW — 1 min read six cards; 2 min organise; 1 min explain a dependency. IF REVISION — 0.5 min read; 2.5 min organise and justify; 1 min check.

Stage	Minutes	Task	Staff move
You do · prepare a valid comparison	8	Use the supplied evidence. Write, draw, point, explain or direct an adult.	Use supplied headings and card letters rather than rewriting a long method. Switch routes at the task with no record. Pointing, annotation or pupil-directed scribing preserve the scientific demand. IF NEW — 2 min use reference; 6 min complete five concise points; 2 min use one precise feedback point. IF REVISION — 1 min use reference; 7 min complete five concise points; 2 min use one precise feedback point.
Review and improve	3	Keep one correct science claim. Improve one word, unit, piece of evidence or reason. Mark a real change //; read it back if asked.	Give one specific VF and one NS+; wait for a real // edit and hear the pupil explain the change.
Exit · mechanism, measure, control	3	Use your book OR one ticket. Show your explanation with the supplied evidence. Say it, write it, draw it, sign it, choose it or show it; an adult can scribe it.	Use a book response OR one ticket. The pupil owns the response; no counter-initial or extra receipt. Use the evidence to choose last lesson's one next teaching move. IF NEW — 1 min read evidence; 2 min respond; 1 min engage against model answer. IF REVISION — 0.5 min orient; 2.5 min respond; 1 min engage against model answer.

Planned adaptations

Need	Planned adaptation	EEF five-a-day strategy
Cognition and learning	Arrival: reveal one clue at a time; model a reasoning step, then fade the cue for a fresh example.	scaffolding
Communication and interaction	We do: allow thinking time, paired rehearsal, pointing, sign or AAC before giving a reason.	flexible grouping

SEMH	Independent: agree a quiet pause and one return step; correct privately; accept a pass on a personal example.	cognitive and metacognitive strategies
Sensory and physical	Models: offer reduced motion, paused diagrams, enlarged prepared pieces, seated work or adult-directed placement.	technology
EAL (language access, not SEN)	Arrival: pre-teach topic words beside the diagram; permit any-language planning and a sentence stem for the same science.	explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below. EEF strategies are linked to actual tasks above.

If behind

Must teach core: I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely. Shorten or merge: take one shared explanation; preserve independent evidence, review and exit. Next lesson: revisit any insecure criterion before extending the topic. Dependency: Use enzyme, substrate, active site and specificity accurately in an explanation.

Staff evaluation

Date taught	Record actual date.
Delivered as planned	Yes / Partly / No; name change.
Criteria met	Name the criterion and evidence.
Adaptations made and why	Name actual support and effect.
Change from the SoW and coverage impact	State change, reason and next teaching implication.
Evidence banked	Record evidence location and lesson code on the school's evidence app.

P18 · Individual target opportunities

Cognition and learning: Arrival: reveal one clue at a time; model a reasoning step, then fade the cue for a fresh example.

Communication and interaction: We do: allow thinking time, paired rehearsal, pointing, sign or AAC before giving a reason.

SEMH: Independent: agree a quiet pause and one return step; correct privately; accept a pass on a personal example.

Sensory and physical: Models: offer reduced motion, paused diagrams, enlarged prepared pieces, seated work or adult-directed placement.

EAL (language access, not SEN): Arrival: pre-teach topic words beside the diagram; permit any-language planning and a sentence stem for the same science.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan.

Follow the pupil's IBSP where it differs from these routines.

Adult deployment and responsive feedback

At Independent, one adult supports access to Supported while the other holds the group; swap across the week. Wait, gesture, cue, model one step, then try a fresh step independently. Give VF, then one precise

NS+; wait for a real pupil response R and // edit. Use ? for read-back. WS and I describe actual support; E requires target-matched evidence.

During We do, the lesson teacher is the named audience. Offer an optional request for reading help, fewer pieces, clearer labels, adult-directed handling or thinking time. Try the change and check its effect. Passing and private requests are valid.

First lesson back · once-only read

Before the lesson, scan the books for five minutes. Notice previous feedback, pupil responses and where the learning stopped.

Ask the cover teacher and TAs what was actually taught, who was present and what changed for individual pupils.

Check the roll against the books. New arrivals have no prior evidence; start from what every pupil shows now.

Existing NS+ is history. Do not chase, back-mark or retrospectively close it. Begin fresh feedback from this lesson.

After the check, one read, once, shapes Autumn 2. Read each success criterion with its access support, choose the next route, then teach; do not reopen earlier feedback.

Preparation and practical safety

Use the supplied models, cards, data and booklet. This lesson plans only. During the later practical, handle reagents only with the centre-approved method, prepared equipment, local risk assessment and trained supervision. The supplied-observation route needs no reagents and teaches the same interpretation; it does not certify physical practical completion. Do not infer concentrations, heating arrangements or disposal from lesson cards.

Destination link (Launch): this planning work is GCSE Biology core-practical ground; the same enzyme and method skills are used in food and drink production and in hospital and diagnostic laboratories, and lead on to Level 3 science and laboratory apprenticeships.

Choose a route at each task using what the pupil shows. Switching adds no task, record or label.

Use paper models or teacher-authorised apparatus. No tasting, personal genetic disclosure, physical contact or unplanned chemical handling. Offer adult-directed handling and a seated route.

Teacher answers and topic guidance

Real-world link

Shared activities

Model: 1 amylase enzyme; 2 starch substrate; 3 active site; 4 smaller sugar products. The enzyme is not used up. Method order F, B, D, A, C, E, allowing F/B swap. Blue-black detects starch; the first orange-brown remaining sample gives the endpoint. No-active-amylase control expected blue-black.

Supported arrival

1. Speeds up a reaction without being used up. Enzyme action is introduced today; this is a supplied starting fact.
2. Starch. Amylase breaks starch into smaller sugars.
3. Change pH only, while keeping temperature and other relevant conditions the same.
4. pH. Time is the measured outcome, or dependent variable.

Supported independent task

1. Amylase: enzyme; starch: substrate; not used up.
2. Change pH; measure endpoint time; keep temperature the same because it also affects activity.
3. F, B, D, A, C, E. F and B may swap if complete before mixing.
4. Endpoint: orange-brown remains. No-active-amylase comparison: blue-black, because starch should remain detectable.
5. Use the supplied four headings. Record "no endpoint by 300 s"; do not invent zero time or zero rate.

Supported exit

1. Starch fits the complementary active site. The enzyme is not used up, so it can catalyse another reaction. Accept correct labelled pointing with explanation or exact scribing.
2. Change pH. Time until a sample leaves iodine orange-brown, in seconds. Keep a relevant factor such as temperature or enzyme/starch volume or concentration the same.

Independent arrival

1. It speeds up the reaction and is not used up, so it can act again.
2. Substrate: starch. Enzyme: amylase.
3. So a change in time or rate is not confused with a temperature effect. It makes the comparison more valid for investigating pH.
4. Independent: pH. Dependent: time to the defined reaction endpoint, measured in seconds.

Independent independent task

1. pH may affect activity, giving different endpoint times; the fastest tested condition is to be found. Products leave and amylase is not used up.
2. IV pH; DV time / s to the first sampled drop leaving iodine orange-brown. Controls include temperature, concentrations, volumes and sampling conditions. Temperature also affects activity.
3. F, B, D, A, C, E (F/B may swap). Separate iodine drops avoid changing the main reaction mixture with the detector.
4. A control variable is kept the same; a control sample is an additional comparison. Without active amylase starch should remain detectable, giving blue-black.
5. Record "no endpoint by 300 s"; leave numerical time/rate uncalculated for that run. 1/time compares matched starting amounts and endpoints; it is not product mass per second.

Independent exit

1. The active site has a complementary fit with the starch substrate, enabling catalysis. Products are released and the enzyme is not used up, so it can act again.
2. IV: pH. DV: time / s until a sample leaves iodine orange-brown. For example, keep temperature constant because it also affects enzyme activity and could confuse the effect of pH.

Aspire arrival

1. Amylase catalyses the reaction and can act again. Starch is the substrate converted into smaller sugar products.
2. Both factors could change the rate, so a difference cannot be attributed to pH alone. Keep amylase concentration constant.
3. Repeats show variation, help identify doubtful runs and allow a more reliable summary of the time at that pH. They do not remove a systematic error by themselves.
4. Otherwise different stopping rules may create apparent time differences. A shared observable endpoint makes the comparison more consistent.

Aspire independent task

1. pH may alter binding/catalysis, changing endpoint time. A complementary active site binds starch; products leave and the catalyst is not used up. Slower activity alone cannot establish permanent denaturation.
2. IV mixture pH; DV time / s to first orange-brown sample. Match temperature, enzyme/starch concentration and volume, total volume, start/mixing and sampling conditions. Link at least two to plausible rate/timing confounds.
3. F, B, D, A, C, E, allowing F/B swap. Separate iodine avoids altering the reaction; a consistent mixing/timer start prevents an artificial time difference.
4. A variable is matched; the control sample lacks active enzyme. Blue-black expected. If orange-brown, check starting starch, reagent function, sample identity and contamination before interpreting enzyme action.
5. Record 120 s; threshold crossed after 90 and by 120 s, not exactly 105 s. A shorter feasible fixed sampling interval reduces interval uncertainty. Record "no endpoint by 300 s"; no invented numerical rate.

Aspire exit

1. A complementary active site enables substrate binding and catalysis; products leave and the enzyme remains available. Unsuitable pH can alter binding or catalysis, sometimes through structural change, reducing activity. Reduced activity alone does not establish permanent denaturation.
2. IV: mixture pH. DV: time / s to the first sampled drop leaving iodine orange-brown. A justified matched factor such as temperature, enzyme/starch concentration or sampling interval reduces a competing cause of time differences.

OpenStax: enzymes

Next teaching after last lesson: Autumn 2, after half term.

LAUNCH SCIENCE · last lesson · Week 8 · Autumn 2

Staff copy — not for pupil books

Lead with VF in the practical and exit tasks. Written feedback is for extended writing only, strategic and not every piece (Curriculum Policy 2026/27 §14.3).

Meaning

Verbal feedback given here

Worked with support

Next step one specific thing

Evidence captured on the school's digital evidence platform

Responded loop closed (Audience has happened)

Self-edit point

Read this back to me

Curriculum Policy 2026/27 §14.3 summary: VF WS I NS E R (six codes; NS “next step”, E “evidence on digital software app”, R “responded to feedback”). The Feedback Policy, the code source, adds // and ? and expands NS to one specific thing and R to loop closed.

Voice + Audience = R: the pupil responds AND the adult genuinely engages with that response; add R only when both have happened. For a margin edit, the policy says: “in the same colour, on the same page”.

Supported: Label enzyme and substrate, then add why the enzyme can act again.

Independent: Replace “measure speed” with the observable iodine endpoint and seconds.

Aspire: Separate a possible pH mechanism from what endpoint timing alone can establish.

Secure: Teach how sampling interval limits endpoint precision.

Mixed: Model only the incomplete link between pH, enzyme activity and time.

Misconception: Contrast enzyme reuse with substrate change, or the two iodine outcomes.

Access barrier: Use enlarged labels and the three-heading planning frame.

Method-or-data problem: Rebuild a matched comparison and a truthful bounded time entry.

We do: label the enzyme model. Invite a pupil to place and explain a label, then engage with their explanation. Use VF when verbal feedback is actually given.

I've been out for a while and haven't seen your work, so today I'll start from what you show me.

LAUNCH: “Adult engages with the edit against the mark scheme / model answer before R.” The pupil owns the response; no adult counter-initial or extra receipt. Codes stay in books. No second next step or retrospective closure.

Answer: Speeds up a reaction without being used up. Enzyme action is introduced today; this is a supplied starting fact.

Answer: Starch. Amylase breaks starch into smaller sugars.

Answer: Change pH only, while keeping temperature and other relevant conditions the same.

Answer: pH. Time is the measured outcome, or dependent variable.

Answer: It speeds up the reaction and is not used up, so it can act again.

Answer: Substrate: starch. Enzyme: amylase.

Answer: So a change in time or rate is not confused with a temperature effect. It makes the comparison more valid for investigating pH.

Answer: Independent: pH. Dependent: time to the defined reaction endpoint, measured in seconds.

Answer: Amylase catalyses the reaction and can act again. Starch is the substrate converted into smaller sugar products.

Answer: Both factors could change the rate, so a difference cannot be attributed to pH alone. Keep amylase concentration constant.

Answer: Repeats show variation, help identify doubtful runs and allow a more reliable summary of the time at that pH. They do not remove a systematic error by themselves.

Answer: Otherwise different stopping rules may create apparent time differences. A shared observable endpoint makes the comparison more consistent.

Additional task and diagram wording

LAUNCH · SCIENCE · DISCOVER

40 minutes · take a pause when you need one. You may pass on a personal example.

Nothing typed here is stored or sent. Use Save my responses to keep a copy.

Point · speak · draw · type · write

Start with the arrival task.

1. Catalyst and substrate

An enzyme is a biological catalyst: it speeds a reaction without being used up. Choose: speeds up / always stops a reaction.

The fact is in the first sentence.

2. Substrate or comparison

Amylase breaks down starch. Which is its substrate here: starch, protein or lipid?

Substrate means the substance the enzyme acts on.

3. Controls or repeats

To check the effect of pH, should we change pH only or change pH and temperature together?

Keep other rate factors the same.

4. Changed factor and endpoint

We change pH and measure a time. Which is the independent variable: pH or time?

Independent means the factor we deliberately change.

Starting fact: an enzyme is a biological catalyst. A catalyst speeds a reaction without being used up.

Describe its job.

Use both parts of the supplied definition.

Starting fact: amylase breaks starch into smaller sugars. Name the substrate and the enzyme.

The substrate is what is acted on.

Why keep temperature the same when testing the effect of pH on a reaction?

Temperature can also affect reaction speed.

An investigation changes pH and times a reaction endpoint. Name the independent and dependent variables.

Changed factor versus measured outcome.

Starting fact: catalysts speed reactions without being used up. How is the role of amylase different from the role of starch?

Amylase is the enzyme; starch is its substrate.

Two trials use different pH values AND different amylase concentrations. Why is this a problem?

Can an effect be attributed to pH alone?

Why repeat a reaction-time measurement at the same pH?

Think about variation and a doubtful result.

Why must everyone agree what counts as the reaction endpoint before starting a timed comparison?

The stopping rule is part of the measurement.

I have answered in my book or out loud.

Starch is not detected in the sampled drop.

These are separate iodine checks on a spotting tile. The colours are supported by words; colour alone is not the answer.

Which observation shows starch is still present? Which might help us decide when the reaction has reached an endpoint?

Read both colour descriptions. Point to the one that still detects starch.

Separate the observed colour from what it tells you about starch.

Define an observable endpoint; avoid claiming what the check does not detect.

Specificity: an enzyme acts on particular substrates because of the complementary active site.

A shape model, not a literal picture of molecules. Starch is a long molecule; we show only a short segment.

Enzyme shapes can flex.

Choose a card, then a destination. You can change your mind.

Find the large purple shape first. Then locate the notch and yellow pieces.

Match each name to its function, then check the numbered part.

Distinguish the enzyme from the substance changed, and explain why the enzyme remains.

At an unsuitable pH, binding and catalysis may be less effective.

Denaturation is a change in an enzyme's three-dimensional structure that disrupts its active site and function.

We will investigate which tested pH gives the shortest endpoint time.

A complementary active site can bind this substrate.

The model does not give a universal optimum pH. Reduced activity alone does not prove permanent denaturation.

Which observation is our endpoint?

pH measures acidity or alkalinity. For matched starting amounts and the same endpoint, shorter time means a higher relative rate.

A · Sample at a fixed interval into separate iodine drops. ↑ ↓

B · Prepare equal mixtures and constant temperature using the approved setup. ↑ ↓

C · Record the first sample where iodine remains orange-brown. ↑ ↓

D · Mix enzyme with starch and start timing as directed. ↑ ↓

E · Repeat at each pH with the same conditions. ↑ ↓

F · Prepare the results headings before starting. ↑ ↓

Use Up and Down to organise the steps.

F and B may swap if both are complete before mixing. Today is a planning lesson; use the teacher's approved method for the later practical.

Sort into before timing, during the reaction and after the endpoint. Then order each group.

Check the start event, repeated sampling and recording rule.

Identify dependencies, matched conditions and any alternative valid order.

Enzyme: biological catalyst; speeds a reaction without being used up. Substrate: substance it acts on. pH: a measure of acidity or alkalinity. Amylase acts on starch, producing smaller sugars.

Iodine key: blue-black = starch detected. Orange-brown remains = starch not detected. check separate sampled drops. Endpoint = first sampled drop leaving iodine orange-brown; record its time / s.

Prepare matched mixtures and temperature using the centre-approved setup.

Mix enzyme and starch; start timing consistently as directed.

Transfer samples at the fixed interval into separate iodine drops.

Repeat at each pH using matched conditions.

Valid comparison: change pH; match temperature, enzyme and starch concentrations/volumes, total volume, starting and sampling conditions. A separate no-active-amylase control should remain starch-positive. This is different from a factor kept constant.

Planning only: use the centre-approved practical method later. Table headings: pH | trial | endpoint time / s | observation. A relative-rate column, $1/\text{time} / \text{s}^{-1}$, needs an observed endpoint and matched starting amount.

Illustrative timing example: blue-black at 90 s; orange-brown at 120 s (30 s sampling). Record 120 s; the detection threshold was crossed after 90 and by 120 s. Another run has no endpoint by 300 s: no numerical rate is established.

1. Use the model to finish: "Amylase is the _____. Starch is the _____. The enzyme is not _____."

Word bank: enzyme · substrate · used up. Trace the purple shape before and after.

2. Match pH / time to endpoint / temperature to CHANGE / MEASURE / KEEP THE SAME. Give one reason for keeping temperature the same.

Independent = change. Dependent = measure. Temperature can also change reaction speed.

3. Order method cards A–F on the reference page. Point to the actions done before mixing.

Find BEFORE (F/B), DURING (D/A), then ENDPOINT and REPEATS (C/E).

4. Point to the endpoint sample. Predict the iodine result in a separate sample with starch but no active amylase.

Blue-black = starch detected; orange-brown remains = not detected. The control still contains starch.

5. Use these results headings: pH | trial | endpoint time / s | observation. If no endpoint is seen by 300 s, choose "0 s" or "no endpoint by 300 s".

The agreed stopping time is a limit on the observation. It is not a measured reaction time.

Write a testable prediction for how pH could affect amylase acting on starch. Explain why the enzyme can act again. [2 marks]

Use may, then link active-site binding to activity. Do not assume a universal optimum pH.

Name the IV and define the timed DV with units. Give three controls and justify one. [6 marks]

Change / measure / keep the same. Define the iodine stopping rule, not simply "how fast".

Order the six method cards and explain why iodine is used to test separate sampled drops. [2 marks]

Group before / during / after. Keep iodine out of the main reaction vessel.

Distinguish a no-active-amylase control sample from a control variable. Predict the control result. [3 marks]

Constant temperature is a factor; a no-active-enzyme mixture is a separate comparison.

State suitable table headings for pH, trial, endpoint time / s, observation and relative rate $1/\text{time} / \text{s}^{-1}$. State what to record if there is no endpoint by 300 s. [3 marks]

A bounded observation is not zero. Calculate a reciprocal only for an observed endpoint time.

Predict a testable pH–activity pattern. Explain specificity and reuse without claiming that reduced activity proves permanent denaturation. [4 marks]

Connect mechanism to a measurable prediction. A shape model illustrates a possibility; it does not establish the pH optimum.

Specify the IV, operational DV and four controls. Justify two controls against competing explanations. [8 marks]

Operational means another person can apply the same stopping rule. Temperature and concentration could alter activity independently of pH.

Order method cards A–F; explain why iodine goes into separate check drops and why a consistent start matters. [3 marks]

Timing must start consistently with mixing. The detector must not become a second changed condition.

Distinguish the no-active-amylase control from a control variable. What would an unexpected orange-brown result make you check? [3 marks]

Use expected starch presence to reason backwards. An unexpected control questions validity before it supports a mechanism.

State the evidence from illustrative samples: blue-black at 90 s, orange-brown at 120 s, sampled every 30 s. State the recorded time, the interval containing the threshold crossing, and one precision improvement. For another run with no endpoint by 300 s, choose a truthful table entry. [4 marks]

The first negative starch sample is an upper sampling bound. Choose headings pH / trial / time or no endpoint by... / observation; explain what your improvement addresses.

LAUNCH · SCIENCE · REVIEW

Plan: investigate how pH affects amylase acting on starch. Iodine turns blue-black when starch is detected; it remains orange-brown when starch is not detected. Today is preparation for the later investigation; no reaction results have been collected.

1. Why can amylase act on starch and then act again? Use the model or complete: ‘Starch fits the _____. The enzyme is not _____.’

Word bank: active site · used up · product. Choose the words that explain binding and reuse.

Answer guidance: Starch fits the complementary active site. The enzyme is not used up, so it can catalyse another reaction. Accept correct labelled pointing with explanation or exact scribing.

2. For the pH check, name what we change, what we time and one thing we keep the same.

Use the headings Change / Time until... / Keep the same. Use the detector key to finish the timing rule.

Answer guidance: Change pH. Time until a sample leaves iodine orange-brown, in seconds. Keep a relevant factor such as temperature or enzyme/starch volume or concentration the same.

1. Explain why amylase acts on starch and can be used again.

Link complementary fit at the active site to specificity; then distinguish enzyme from substrate.

Answer guidance: The active site has a complementary fit with the starch substrate, enabling catalysis.

Products are released and the enzyme is not used up, so it can act again.

2. Name the IV, define the timed DV with units, and name one control with a reason.

The DV needs an observable endpoint. Explain how changing another rate factor would affect a comparison of pH.

Answer guidance: IV: pH. DV: time / s until a sample leaves iodine orange-brown. For example, keep temperature constant because it also affects enzyme activity and could confuse the effect of pH.

1. Explain enzyme specificity and reuse. Predict how an unsuitable pH could affect this mechanism.

Link active-site shape, binding and product release. Do not assume that every reduction in activity proves permanent denaturation.

Answer guidance: A complementary active site enables substrate binding and catalysis; products leave and the enzyme remains available. Unsuitable pH can alter binding or catalysis, sometimes through structural change, reducing activity. Reduced activity alone does not establish permanent denaturation.

2. Specify the IV, endpoint-time DV with units, and one justified control for a valid pH comparison.

Make the stopping rule observable and show how the chosen control prevents a competing explanation.

Answer guidance: IV: mixture pH. DV: time / s to the first sampled drop leaving iodine orange-brown. A justified matched factor such as temperature, enzyme/starch concentration or sampling interval reduces a competing cause of time differences.

LAUNCH · GCSE Biology · Enzyme Action: Preparing the pH Investigation

Blue-black iodine spot: starch detected

Orange-brown iodine spot: starch not detected

Amylase enzyme with complementary active site, starch substrate and smaller sugar products

Complete staff planning reference

Independent arrival

Independent independent task

Independent exit

Aspire arrival

Aspire independent task

Aspire exit

Independent: Replace “measure speed” with the observable iodine endpoint and seconds.

Aspire: Separate a possible pH mechanism from what endpoint timing alone can establish.

2. Name the IV and define the timed DV with units. Give three controls and justify one.

3. Order the six method cards and explain why iodine check separate sampled drops.

4. Distinguish a no-active-amylase control sample from a control variable. Predict the control result.

5. Use table headings pH | trial | endpoint time / s | observation | relative rate $1/\text{time} / \text{s}^{-1}$. State what to record if there is no endpoint by 300 s.

2. Specify the IV, operational DV and four controls. Justify two controls against competing explanations.

3. Order method cards A–F; explain why iodine goes into separate check drops and why a consistent start matters.

4. Distinguish the no-active-amylase control from a control variable. What would an unexpected orange-brown result make you check?

5. Illustrative samples: blue-black at 90 s, orange-brown at 120 s, sampled every 30 s. State the recorded time, the interval containing the threshold crossing, and one precision improvement. For another run with no endpoint by 300 s, choose a truthful table entry.

Current science lesson guidance and mark scheme

Science staff guidance

Beside the You do, mark I (independent), S (supported) or C (choice): the level of independence, not the effort.

Give time to start → point to the relevant place → rephrase → offer a small prompt or word bank → reduce to a limited choice. Never straight to telling the answer.

No cold calling; nobody is asked to read difficult text aloud.

Tell the teacher the barrier: reading, vocabulary, processing, regulation, writing or a science misconception. Avoid prompting through every question or dictating the final answer.

E Evidence on the school's evidence app

Lean capture on the school's evidence app (E): 30-second clip, photo, or one-line context note.

Point, sign, repeat after the adult, demonstrate, a clip on the school's evidence app, choice board.

If behind

A pupil who missed last lesson: give the Arrival reminders and today's must-know words, not the missed lesson's work.

Arrival and Starter

Arrival: answer these four short questions in the book.

1. Which structure holds chromosomes in a eukaryotic animal cell?
2. In diffusion, do particles move from higher to lower concentration or the reverse?
3. Which substance moves through a partially permeable membrane in osmosis?
4. Which transport process uses energy to move particles against a concentration gradient?

Starter: Which iodine observation shows starch is still present: blue-black or orange-brown? One short answer in the book.

You do routes

Independent is preferred: work in the book without answer-giving support. Supported uses a small prompt beside the same task. Choice uses limited options, matching or selecting and then a short scientific sentence, label or reason in the book. Aspire is the challenge for every route.

Mark scheme · one point per mark

Independent

1. Write a testable prediction for how pH could affect amylase acting on starch. Explain why the enzyme can act again. [2 marks]

1. Changing pH may change endpoint time under matched conditions (1 mark)
2. Amylase is not used up and can catalyse another reaction (1 mark)

2. Name the IV and define the timed DV with units. Give three controls and justify one. [6 marks]

1. pH is deliberately changed (1 mark)
2. Time to first orange-brown iodine sample is measured in seconds (1 mark)
3. Keep temperature constant (1 mark)
4. Keep enzyme concentration or volume constant (1 mark)
5. Keep starch concentration or volume constant (1 mark)
6. A named control could otherwise alter rate, so matching it isolates pH (1 mark)

3. Order the six method cards and explain why iodine is used to test separate sampled drops. [2 marks]

1. F/B before D, then A, C, E, with preparation before mixing (1 mark)
2. Sampling checks starch without adding iodine to the main reaction (1 mark)

4. Distinguish a no-active-amylase control sample from a control variable. Predict the control result. [3 marks]

1. A control variable is a factor kept constant (1 mark)
2. The no-active-amylase sample is a separate comparison (1 mark)
3. It should stay starch-positive and turn iodine blue-black (1 mark)

5. State suitable table headings for pH, trial, endpoint time / s, observation and relative rate $1/\text{time} / \text{s}^{-1}$. State what to record if there is no endpoint by 300 s. [3 marks]

1. Headings include pH, trial, endpoint time / s, observation and relative rate / s^{-1} (1 mark)

2. Record no endpoint by 300 s (1 mark)
3. Do not calculate an exact reciprocal rate for an unobserved endpoint (1 mark)

Aspire

1. Predict a testable pH–activity pattern. Explain specificity and reuse without claiming that reduced activity proves permanent denaturation. [4 marks]
 1. A testable prediction relates changed pH to endpoint time (1 mark)
 2. Substrate fits a complementary active site (1 mark)
 3. Enzyme is not used up (1 mark)
 4. Slower activity alone does not establish permanent denaturation (1 mark)
2. Specify the IV, operational DV and four controls. Justify two controls against competing explanations. [8 marks]
 1. pH is the independent variable (1 mark)
 2. The measured variable is the first starch-negative sampled time in seconds (1 mark)
 3. Match temperature (1 mark)
 4. Match amylase concentration and volume (1 mark)
 5. Match starch concentration and volume (1 mark)
 6. Match total volume, starting conditions or sampling interval (1 mark)
 7. Justify one control as excluding an alternative rate effect (1 mark)
 8. Justify a second control as excluding an alternative rate effect (1 mark)
3. Order method cards A–F; explain why iodine goes into separate check drops and why a consistent start matters. [3 marks]
 1. F/B before D, then A, C, E (1 mark)
 2. Iodine is kept out of the reacting mixture (1 mark)
 3. A consistent mixing/start event keeps the timing comparison valid (1 mark)
4. Distinguish the no-active-amylase control from a control variable. What would an unexpected orange-brown result make you check? [3 marks]
 1. Matched conditions are controlled variables (1 mark)
 2. The no-active-amylase control is a separate comparison (1 mark)
 3. Orange-brown would prompt a check for absent starch, contamination or an inactive detector before accepting results (1 mark)
5. State the evidence from illustrative samples: blue-black at 90 s, orange-brown at 120 s, sampled every 30 s. State the recorded time, the interval containing the threshold crossing, and one precision improvement. For another run with no endpoint by 300 s, choose a truthful table entry. [4 marks]
 1. Record 120 s (1 mark)
 2. Crossing occurred after 90 and by 120 s (1 mark)
 3. A shorter authorised sampling interval narrows the timing window (1 mark)
 4. Record no endpoint by 300 s for the other run, without an invented exact rate (1 mark)

Accept scientifically equivalent wording. Award each distinct point once. The mark tariff is for the question, not a participation score. Keep the answer scheme staff-only during independent work.

Arrival 1: Which structure holds chromosomes in a eukaryotic animal cell?

Arrival 2: In diffusion, do particles move from higher to lower concentration or the reverse?

Arrival 3: Which substance moves through a partially permeable membrane in osmosis?

Arrival 4: Which transport process uses energy to move particles against a concentration gradient?

Starter: Which iodine observation shows starch is still present: blue-black or orange-brown?

Under the heading Exit ticket: Explain why amylase acts on starch and can be used again. Name the IV, define the timed DV with units, and name one control with a reason.